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Last login: Wed Jun 17 10:30:38 on ttys003
stephenrajva@Stephens-MacBook-Air-4 ~ % cd ~/Downloads/HPC-Upload\ 2/miniapps/MiniFE_simulation/dragonfly/dfly4
stephenrajva@Stephens-MacBook-Air-4 dfly4 % ls
dfly_hopcount.py                                hopcount_dragonfly.csv
dfly_minife_hopcount.png                        minife_communication_stats_dragonfly.csv
dfly4.py
stephenrajva@Stephens-MacBook-Air-4 dfly4 % grep "send_bit_count" minife_communication_stats_dragonfly.csv | head
rtr_G0R0, send_bit_count, port0, Accumulator, 1210569133, 0, 781032, 2339823680, 607, 360, 4280
rtr_G0R0, send_bit_count, port1, Accumulator, 1210569133, 0, 206192, 493517568, 235, 360, 3632
rtr_G0R0, send_bit_count, port2, Accumulator, 1210569133, 0, 576120, 1847964992, 372, 360, 4280
nic0:rtrLink:networkIF, send_bit_count, , Accumulator, 1210569133, 0, 782312, 2341482560, 607, 360, 4280
rtr_G0R1, send_bit_count, port0, Accumulator, 1210569133, 0, 283832, 524407360, 432, 360, 3632
rtr_G0R1, send_bit_count, port1, Accumulator, 1210569133, 0, 540712, 1811105088, 340, 360, 4280
rtr_G0R1, send_bit_count, port2, Accumulator, 1210569133, 0, 80728, 32731840, 203, 360, 808
nic1:rtrLink:networkIF, send_bit_count, , Accumulator, 1210569133, 0, 283832, 524407360, 432, 360, 3632
rtr_G1R0, send_bit_count, port0, Accumulator, 1210569133, 0, 243408, 530560640, 273, 360, 3632
rtr_G1R0, send_bit_count, port1, Accumulator, 1210569133, 0, 337608, 1319429568, 111, 360, 4280
stephenrajva@Stephens-MacBook-Air-4 dfly4 % grep "send_packet_count" minife_communication_stats_dragonfly.csv | head
rtr_G0R0, send_packet_count, port0, Accumulator, 1210569133, 0, 607, 607, 607, 1, 1
rtr_G0R0, send_packet_count, port1, Accumulator, 1210569133, 0, 235, 235, 235, 1, 1
rtr_G0R0, send_packet_count, port2, Accumulator, 1210569133, 0, 372, 372, 372, 1, 1
rtr_G0R1, send_packet_count, port0, Accumulator, 1210569133, 0, 432, 432, 432, 1, 1
rtr_G0R1, send_packet_count, port1, Accumulator, 1210569133, 0, 340, 340, 340, 1, 1
rtr_G0R1, send_packet_count, port2, Accumulator, 1210569133, 0, 203, 203, 203, 1, 1
rtr_G1R0, send_packet_count, port0, Accumulator, 1210569133, 0, 273, 273, 273, 1, 1
rtr_G1R0, send_packet_count, port1, Accumulator, 1210569133, 0, 111, 111, 111, 1, 1
rtr_G1R0, send_packet_count, port2, Accumulator, 1210569133, 0, 267, 267, 267, 1, 1
rtr_G1R1, send_packet_count, port0, Accumulator, 1210569133, 0, 308, 308, 308, 1, 1
stephenrajva@Stephens-MacBook-Air-4 dfly4 % grep "output_port_stalls" minife_communication_stats_dragonfly.csv | head
rtr_G0R0, output_port_stalls, port0, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port1, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port2, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
nic0:rtrLink:networkIF, output_port_stalls, , Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_G0R1, output_port_stalls, port0, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_G0R1, output_port_stalls, port1, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_G0R1, output_port_stalls, port2, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
nic1:rtrLink:networkIF, output_port_stalls, , Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_G1R0, output_port_stalls, port0, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_G1R0, output_port_stalls, port1, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
stephenrajva@Stephens-MacBook-Air-4 dfly4 % grep "idle_time" minife_communication_stats_dragonfly.csv | head
rtr_G0R0, idle_time, port0, Accumulator, 1210569133, 0, 1202093919, 2684553836409781, 608, 126401, 16329868
rtr_G0R0, idle_time, port1, Accumulator, 1210569133, 0, 1208279039, 138756371633261109, 236, 126401, 367417343
rtr_G0R0, idle_time, port2, Accumulator, 1210569133, 0, 1204368017, 9229280865874509, 373, 1417401, 69578401
nic0:rtrLink:networkIF, idle_time, , Accumulator, 1210569133, 0, 545289457, 790386443568297, 607, 120669, 12178536
rtr_G0R1, idle_time, port0, Accumulator, 1210569133, 0, 1207357936, 135517701489085740, 433, 47735, 365558143
rtr_G0R1, idle_time, port1, Accumulator, 1210569133, 0, 1204741257, 14101092471062875, 341, 126401, 71413336
rtr_G0R1, idle_time, port2, Accumulator, 1210569133, 0, 1209612039, 192860267370926469, 204, 125068, 435184801
nic1:rtrLink:networkIF, idle_time, , Accumulator, 1210569133, 0, 892066543, 133915381360087381, 432, 124668, 364766810
rtr_G1R0, idle_time, port0, Accumulator, 1210569133, 0, 1207885804, 38214821769304744, 274, 125068, 153710936
rtr_G1R0, idle_time, port1, Accumulator, 1210569133, 0, 1206995360, 422069244149460004, 112, 125601, 521869868
stephenrajva@Stephens-MacBook-Air-4 dfly4 % grep "latency" minife_communication_stats_dragonfly.csv | head -20
nic0:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 129941, 28499721, 607, 198, 285
nic1:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 85625, 16971461, 432, 198, 199
nic2:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 54550, 10943882, 273, 197, 285
nic3:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 70158, 16502522, 308, 197, 285
stephenrajva@Stephens-MacBook-Air-4 dfly4 %
stephenrajva@Stephens-MacBook-Air-4 dfly4 % cd ~/Downloads/HPC-Upload\ 2/miniapps/MiniFE_simulation/dragonfly/dfly16
stephenrajva@Stephens-MacBook-Air-4 dfly16 % ls
dfly16_hopcount.py                                dfly16.py                                hopcount_dragonfly.csv minife_dfly16.csv
stephenrajva@Stephens-MacBook-Air-4 dfly16 % grep "send_bit_count" minife_dfly16.csv | head
rtr_G0R0, send_bit_count, port0, Accumulator, 2012790733, 0, 786648, 2345301440, 613, 360, 4280
rtr_G0R0, send_bit_count, port1, Accumulator, 2012790733, 0, 282176, 523493248, 429, 360, 3632
rtr_G0R0, send_bit_count, port2, Accumulator, 2012790733, 0, 320328, 563340864, 464, 360, 3632
rtr_G0R0, send_bit_count, port3, Accumulator, 2012790733, 0, 411144, 1348392640, 299, 360, 4280
rtr_G0R0, send_bit_count, port4, Accumulator, 2012790733, 0, 159368, 65845568, 397, 360, 808
rtr_G0R0, send_bit_count, port5, Accumulator, 2012790733, 0, 6768, 7634304, 6, 1128, 1128
nic0:rtrLink:networkIF, send_bit_count, , Accumulator, 2012790733, 0, 787928, 2346960320, 613, 360, 4280
rtr_G0R1, send_bit_count, port0, Accumulator, 2012790733, 0, 365432, 559860800, 632, 360, 3632
rtr_G0R1, send_bit_count, port1, Accumulator, 2012790733, 0, 206992, 494250240, 235, 360, 3632
rtr_G0R1, send_bit_count, port2, Accumulator, 2012790733, 0, 84048, 36466048, 206, 360, 1128
stephenrajva@Stephens-MacBook-Air-4 dfly16 % grep "send_packet_count" minife_dfly16.csv | head
rtr_G0R0, send_packet_count, port0, Accumulator, 2012790733, 0, 613, 613, 613, 1, 1
rtr_G0R0, send_packet_count, port1, Accumulator, 2012790733, 0, 429, 429, 429, 1, 1
rtr_G0R0, send_packet_count, port2, Accumulator, 2012790733, 0, 464, 464, 464, 1, 1
rtr_G0R0, send_packet_count, port3, Accumulator, 2012790733, 0, 299, 299, 299, 1, 1
rtr_G0R0, send_packet_count, port4, Accumulator, 2012790733, 0, 397, 397, 397, 1, 1
rtr_G0R0, send_packet_count, port5, Accumulator, 2012790733, 0, 6, 6, 6, 1, 1
rtr_G0R1, send_packet_count, port0, Accumulator, 2012790733, 0, 632, 632, 632, 1, 1
rtr_G0R1, send_packet_count, port1, Accumulator, 2012790733, 0, 235, 235, 235, 1, 1
rtr_G0R1, send_packet_count, port2, Accumulator, 2012790733, 0, 206, 206, 206, 1, 1
rtr_G0R1, send_packet_count, port3, Accumulator, 2012790733, 0, 391, 391, 391, 1, 1
stephenrajva@Stephens-MacBook-Air-4 dfly16 % grep "output_port_stalls" minife_dfly16.csv | head
rtr_G0R0, output_port_stalls, port0, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port1, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port2, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port3, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port4, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port5, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
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nic0:rtrLink:networkIF, output_port_stalls, , Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R1, output_port_stalls, port0, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R1, output_port_stalls, port1, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
rtr_G0R1, output_port_stalls, port2, Accumulator, 2012790733, 0, 0, 0, 0, 0, 0
stephenrajva@Stephens-MacBook-Air-4 dfly16 % grep "idle_time" minife_dfly16.csv | head
rtr_G0R0, idle_time, port0, Accumulator, 2012790733, 0, 2004255534, 11059640529168220, 614, 126935, 21759468
rtr_G0R0, idle_time, port1, Accumulator, 2012790733, 0, 2009599531, 129507911300036665, 430, 22943, 350010143
rtr_G0R0, idle_time, port2, Accumulator, 2012790733, 0, 2009198298, 36863792179365652, 465, 3735, 142351736
rtr_G0R0, idle_time, port3, Accumulator, 2012790733, 0, 2008343845, 37584512330346485, 300, 703733, 135872001
rtr_G0R0, idle_time, port4, Accumulator, 2012790733, 0, 2010904538, 187676823810724448, 398, 121069, 417777601
rtr_G0R0, idle_time, port5, Accumulator, 2012790733, 0, 2012718751, 2092644832873209343, 7, 379203, 1219685403
nic0:rtrLink:networkIF, idle_time, , Accumulator, 2012790733, 0, 1474025606, 7880486546017542, 614, 18533, 17608136
rtr_G0R1, idle_time, port0, Accumulator, 2012790733, 0, 2008618443, 127013810928154135, 633, 22943, 348150943
rtr_G0R1, idle_time, port1, Accumulator, 2012790733, 0, 2010492641, 134490054950821647, 236, 306135, 348090943
rtr_G0R1, idle_time, port2, Accumulator, 2012790733, 0, 2011801647, 186765786491634759, 207, 385068, 416707001
stephenrajva@Stephens-MacBook-Air-4 dfly16 % grep "latency" minife_dfly16.csv | head -20
nic0:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 123239, 25063093, 613, 197, 372
nic1:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 142931, 33397627, 632, 198, 371
nic2:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 183456, 53931802, 667, 197, 371
nic3:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 190235, 55195021, 702, 198, 371
nic4:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 186814, 59498140, 597, 198, 372
nic5:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 186177, 62055001, 597, 198, 377
nic6:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 219933, 81287733, 597, 198, 376
nic7:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 130906, 43408016, 403, 198, 371
nic8:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 76203, 27977633, 209, 198, 372
nic9:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 59517, 17038497, 209, 198, 371
nic10:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 59218, 16890468, 209, 198, 371
nic11:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 75981, 27879951, 209, 198, 371
nic12:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 76629, 28158965, 209, 284, 371
nic13:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 76244, 28007324, 209, 198, 371
nic14:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 75942, 27851398, 209, 197, 371
nic15:rtrLink:networkIF, packet_latency, , Accumulator, 2012790733, 0, 58385, 16394563, 209, 197, 285
stephenrajva@Stephens-MacBook-Air-4 dfly16 %
stephenrajva@Stephens-MacBook-Air-4 dfly16 % cd ~/Downloads/HPC-Upload\ 2/miniapps/MiniFE_simulation/dragonfly/dfly64
stephenrajva@Stephens-MacBook-Air-4 dfly64 % ls
dfly64.py minife_dfly16.csv
stephenrajva@Stephens-MacBook-Air-4 dfly64 % nano dfly64.py
stephenrajva@Stephens-MacBook-Air-4 dfly64 % rm -f minife_dfly64.csv
stephenrajva@Stephens-MacBook-Air-4 dfly64 % sst dfly64.py
File "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniFE_simulation/dragonfly/dfly64/dfly64.py", line 79
    "separator": ", "
    ^
SyntaxError: invalid syntax
FATAL: # SSTPythonModel: Execution of model construction function failed
SST Fatal Backtrace Information:
 0 : 0 sstsim.x 0x0000000104e613c0 _ZNK3SST6Output5fatalEjPKcS2_iS2_z + 644
 1 : 1 sstsim.x 0x0000000104ffff0c4 _ZN3SST4Core24SSTPythonModelDefinition17createConfigG
raphEv + 504
 2 : 2 sstsim.x 0x0000000104dc1524 main + 2692
 3 : 3 dyld 0x0000000106211d54 start + 7184
-----
MPI_ABORT was invoked on rank 0 in communicator MPI_COMM_WORLD
Proc: [[47911,0],0]
Errorcode: 1

NOTE: invoking MPI_ABORT causes Open MPI to kill all MPI processes.
You may or may not see output from other processes, depending on
exactly when Open MPI kills them.
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stephenrajva@Stephens-MacBook-Air-4 dfly64 % nano dfly64.py
stephenrajva@Stephens-MacBook-Air-4 dfly64 % nl -ba dfly64.py | sed -n '70,90p'
 70 system.allocateNodes(ep, "linear")
 71
 72 # Build the system
 73 system.build()
 74
 75 # Statistics configuration
 76 sst.setStatisticLoadLevel(10)
 77 sst.setStatisticOutput("sst.statOutputCSV", {
 78     "filepath": "minife_dfly64.csv"
 79     "separator": ", "
 80 })
 81
 82 # Enable detailed statistics
 83 sst.enableAllStatisticsForComponentType("merlin.hr_router")
 84 sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
 85 sst.enableAllStatisticsForComponentType("ember.EmberEngine", {
 86     "MotifLog": "1"
 87 })
 88 sst.enableAllStatisticsForComponentType("ember.nic")
stephenrajva@Stephens-MacBook-Air-4 dfly64 % nano dfly64.py
stephenrajva@Stephens-MacBook-Air-4 dfly64 % rm -f minife_dfly64.csv
stephenrajva@Stephens-MacBook-Air-4 dfly64 % sst dfly64.py
Allreduce: ranks 64, loop 92, 1 double(s), latency 11.404 us
generate
1052635933:0:0:EmberEngine:Reduce:generate:37: rank=0 size=64
generate
1053979133:0:1:EmberEngine:Reduce:generate:37: rank=1 size=64

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generate
1054108733:0:2:EmberEngine:Reduce:generate:37: rank=2 size=64
generate
1054822333:0:3:EmberEngine:Reduce:generate:37: rank=3 size=64
generate
1055038333:0:5:EmberEngine:Reduce:generate:37: rank=5 size=64
generate
1055047733:0:4:EmberEngine:Reduce:generate:37: rank=4 size=64
generate
1055168533:0:6:EmberEngine:Reduce:generate:37: rank=6 size=64
generate
1055751933:0:7:EmberEngine:Reduce:generate:37: rank=7 size=64
generate
1055882133:0:8:EmberEngine:Reduce:generate:37: rank=8 size=64
generate
1055967933:0:11:EmberEngine:Reduce:generate:37: rank=11 size=64
generate
1055977333:0:9:EmberEngine:Reduce:generate:37: rank=9 size=64
generate
1056098133:0:13:EmberEngine:Reduce:generate:37: rank=13 size=64
generate
1056107533:0:10:EmberEngine:Reduce:generate:37: rank=10 size=64
generate
1056107733:0:12:EmberEngine:Reduce:generate:37: rank=12 size=64
generate
1056228333:0:14:EmberEngine:Reduce:generate:37: rank=14 size=64
generate
1056681533:0:15:EmberEngine:Reduce:generate:37: rank=15 size=64
generate
1056898133:0:16:EmberEngine:Reduce:generate:37: rank=16 size=64
generate
1056898133:0:17:EmberEngine:Reduce:generate:37: rank=17 size=64
generate
1056983933:0:23:EmberEngine:Reduce:generate:37: rank=23 size=64
generate
1057028333:0:18:EmberEngine:Reduce:generate:37: rank=18 size=64
generate
1057078933:0:19:EmberEngine:Reduce:generate:37: rank=19 size=64
generate
1057133333:0:20:EmberEngine:Reduce:generate:37: rank=20 size=64
generate
1057200533:0:27:EmberEngine:Reduce:generate:37: rank=27 size=64
generate
1057210133:0:24:EmberEngine:Reduce:generate:37: rank=24 size=64
generate
1057218933:0:25:EmberEngine:Reduce:generate:37: rank=25 size=64
generate
1057219533:0:21:EmberEngine:Reduce:generate:37: rank=21 size=64
generate
1057330733:0:28:EmberEngine:Reduce:generate:37: rank=28 size=64
generate
1057330733:0:29:EmberEngine:Reduce:generate:37: rank=29 size=64
generate
1057340333:0:22:EmberEngine:Reduce:generate:37: rank=22 size=64
generate
1057349933:0:26:EmberEngine:Reduce:generate:37: rank=26 size=64
generate
1057460333:0:30:EmberEngine:Reduce:generate:37: rank=30 size=64
generate
1057567533:0:33:EmberEngine:Reduce:generate:37: rank=33 size=64
generate
1057653533:0:31:EmberEngine:Reduce:generate:37: rank=31 size=64
generate
1057653933:0:32:EmberEngine:Reduce:generate:37: rank=32 size=64
generate
1057707533:0:34:EmberEngine:Reduce:generate:37: rank=34 size=64
generate
1057740333:0:35:EmberEngine:Reduce:generate:37: rank=35 size=64
generate
1057784333:0:37:EmberEngine:Reduce:generate:37: rank=37 size=64
generate
1057825933:0:47:EmberEngine:Reduce:generate:37: rank=47 size=64
generate
1057870733:0:36:EmberEngine:Reduce:generate:37: rank=36 size=64
generate
1057913933:0:38:EmberEngine:Reduce:generate:37: rank=38 size=64
generate
1057921133:0:39:EmberEngine:Reduce:generate:37: rank=39 size=64
generate
1057956333:0:48:EmberEngine:Reduce:generate:37: rank=48 size=64
generate
1057965133:0:40:EmberEngine:Reduce:generate:37: rank=40 size=64
generate
1057965933:0:49:EmberEngine:Reduce:generate:37: rank=49 size=64
generate
1057974733:0:43:EmberEngine:Reduce:generate:37: rank=43 size=64
generate
1057975533:0:41:EmberEngine:Reduce:generate:37: rank=41 size=64

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generate
1058000333:0:59:EmberEngine:Reduce:generate:37: rank=59 size=64
generate
1058042733:0:55:EmberEngine:Reduce:generate:37: rank=55 size=64
generate
1058061133:0:51:EmberEngine:Reduce:generate:37: rank=51 size=64
generate
1058086733:0:56:EmberEngine:Reduce:generate:37: rank=56 size=64
generate
1058096333:0:50:EmberEngine:Reduce:generate:37: rank=50 size=64
generate
1058105933:0:42:EmberEngine:Reduce:generate:37: rank=42 size=64
generate
1058173133:0:57:EmberEngine:Reduce:generate:37: rank=57 size=64
generate
1058182733:0:45:EmberEngine:Reduce:generate:37: rank=45 size=64
generate
1058191533:0:44:EmberEngine:Reduce:generate:37: rank=44 size=64
generate
1058191533:0:52:EmberEngine:Reduce:generate:37: rank=52 size=64
generate
1058192333:0:53:EmberEngine:Reduce:generate:37: rank=53 size=64
generate
1058216333:0:60:EmberEngine:Reduce:generate:37: rank=60 size=64
generate
1058302733:0:61:EmberEngine:Reduce:generate:37: rank=61 size=64
generate
1058312333:0:46:EmberEngine:Reduce:generate:37: rank=46 size=64
generate
1058312333:0:58:EmberEngine:Reduce:generate:37: rank=58 size=64
generate
1058321933:0:54:EmberEngine:Reduce:generate:37: rank=54 size=64
generate
1058432333:0:62:EmberEngine:Reduce:generate:37: rank=62 size=64
generate
Bcast: ranks 64, loop 2, bytes 56, latency 1.893 us
1058539533:0:63:EmberEngine:Reduce:generate:37: rank=63 size=64
Allgather: ranks 64, loop 3, bytes 12, latency 7.764 us
Barrier: ranks 64, loop 99, latency 11.429 us
2243705933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=1 messageSize=409 iterations=35
2244549133:0:1:EmberEngine:PingPong:generate:65: rank=1 size=64
2244678733:0:2:EmberEngine:PingPong:generate:65: rank=2 size=64
2245392333:0:3:EmberEngine:PingPong:generate:65: rank=3 size=64
PingPong: otherRank 1, total time 63.474 us, loop 35, buflen 409, latency 0.907 us. bandwidth 0.451051 GB/s
2307179533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=2 messageSize=409 iterations=35
2369690333:0:2:EmberEngine:PingPong:generate:65: rank=2 size=64
PingPong: otherRank 2, total time 63.224 us, loop 35, buflen 409, latency 0.903 us. bandwidth 0.452834 GB/s
2370403533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=3 messageSize=82 iterations=35
2432914333:0:3:EmberEngine:PingPong:generate:65: rank=3 size=64
PingPong: otherRank 3, total time 63.224 us, loop 35, buflen 82, latency 0.903 us. bandwidth 0.090788 GB/s
2433627533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=2 messageSize=82 iterations=35
PingPong: otherRank 2, total time 63.224 us, loop 35, buflen 82, latency 0.903 us. bandwidth 0.090788 GB/s
2496851533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=3 messageSize=490 iterations=35
2559362333:0:3:EmberEngine:PingPong:generate:65: rank=3 size=64
PingPong: otherRank 3, total time 63.224 us, loop 35, buflen 490, latency 0.903 us. bandwidth 0.542516 GB/s
2560075533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=3 messageSize=490 iterations=35
PingPong: otherRank 3, total time 63.224 us, loop 35, buflen 490, latency 0.903 us. bandwidth 0.542516 GB/s
Simulation is complete, simulated time: 2.63065 ms
stephenrajva@Stephens-MacBook-Air-4 dfly64 %
stephenrajva@Stephens-MacBook-Air-4 dfly64 % ls *.csv
minife_dfly16.csv          minife_dfly64.csv
stephenrajva@Stephens-MacBook-Air-4 dfly64 % grep "send_bit_count" minife_dfly64.csv | head
rtr_G0R0, send_bit_count, port0, Accumulator, 2630646733, 0, 802632, 2391421760, 619, 360, 4280
rtr_G0R0, send_bit_count, port1, Accumulator, 2630646733, 0, 381416, 605981120, 638, 360, 3632
rtr_G0R0, send_bit_count, port2, Accumulator, 2630646733, 0, 416976, 642110080, 673, 360, 3632
rtr_G0R0, send_bit_count, port3, Accumulator, 2630646733, 0, 589456, 1462698240, 708, 360, 4280
rtr_G0R0, send_bit_count, port4, Accumulator, 2630646733, 0, 429136, 254920064, 1006, 360, 3432
rtr_G0R0, send_bit_count, port5, Accumulator, 2630646733, 0, 435280, 307319936, 994, 360, 3432
rtr_G0R0, send_bit_count, port6, Accumulator, 2630646733, 0, 324528, 190574464, 782, 360, 3432
rtr_G0R0, send_bit_count, port7, Accumulator, 2630646733, 0, 678152, 352220480, 1597, 360, 1896
rtr_G0R0, send_bit_count, port8, Accumulator, 2630646733, 0, 82368, 282686976, 24, 3432, 3432
nic0:rtrlink:networkIF, send_bit_count, , Accumulator, 2630646733, 0, 803912, 2393080640, 619, 360, 4280
stephenrajva@Stephens-MacBook-Air-4 dfly64 % grep "send_packet_count" minife_dfly64.csv | head
rtr_G0R0, send_packet_count, port0, Accumulator, 2630646733, 0, 619, 619, 619, 1, 1
rtr_G0R0, send_packet_count, port1, Accumulator, 2630646733, 0, 638, 638, 638, 1, 1
rtr_G0R0, send_packet_count, port2, Accumulator, 2630646733, 0, 673, 673, 673, 1, 1
rtr_G0R0, send_packet_count, port3, Accumulator, 2630646733, 0, 708, 708, 708, 1, 1
rtr_G0R0, send_packet_count, port4, Accumulator, 2630646733, 0, 1006, 1006, 1006, 1, 1
rtr_G0R0, send_packet_count, port5, Accumulator, 2630646733, 0, 994, 994, 994, 1, 1
rtr_G0R0, send_packet_count, port6, Accumulator, 2630646733, 0, 782, 782, 782, 1, 1
rtr_G0R0, send_packet_count, port7, Accumulator, 2630646733, 0, 1597, 1597, 1597, 1, 1
rtr_G0R0, send_packet_count, port8, Accumulator, 2630646733, 0, 24, 24, 24, 1, 1

```

```

rtr_G0R1, send_packet_count, port0, Accumulator, 2630646733, 0, 603, 603, 603, 1, 1
stephenrajva@Stephens-MacBook-Air-4 dfly64 % grep "output_port_stalls" minife_dfly64.csv | head
rtr_G0R0, output_port_stalls, port0, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port1, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port2, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port3, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port4, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port5, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port6, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port7, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
rtr_G0R0, output_port_stalls, port8, Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
nic0:rtrLink:networkIF, output_port_stalls, , Accumulator, 2630646733, 0, 0, 0, 0, 0, 0
stephenrajva@Stephens-MacBook-Air-4 dfly64 % grep "idle_time" minife_dfly64.csv | head
rtr_G0R0, idle_time, port0, Accumulator, 2630646733, 0, 2621943576, 24621248752246938, 620, 20005, 25737868
rtr_G0R0, idle_time, port1, Accumulator, 2630646733, 0, 2626306485, 118728860919450453, 639, 109335, 317824543
rtr_G0R0, idle_time, port2, Accumulator, 2630646733, 0, 2625933245, 41173883854355443, 673, 49868, 129996536
rtr_G0R0, idle_time, port3, Accumulator, 2630646733, 0, 2624113700, 30915101363181578, 709, 12803, 118155201
rtr_G0R0, idle_time, port4, Accumulator, 2630646733, 0, 2625609326, 153255853873767750, 811, 20009, 376204801
rtr_G0R0, idle_time, port5, Accumulator, 2630646733, 0, 2625557339, 154029242363653241, 994, 12268, 376464001
rtr_G0R0, idle_time, port6, Accumulator, 2630646733, 0, 2626826355, 162980321078743361, 782, 18668, 379964601
rtr_G0R0, idle_time, port7, Accumulator, 2630646733, 0, 2622682058, 169122046771009580, 1009, 56010, 381057601
rtr_G0R0, idle_time, port8, Accumulator, 2630646733, 0, 2629782949, 3494411514791734767, 19, 14418, 1517281409
nic0:rtrLink:networkIF, idle_time, , Accumulator, 2630646733, 0, 2200830449, 20297612428888809, 619, 110005, 21586536
stephenrajva@Stephens-MacBook-Air-4 dfly64 % grep "latency" minife_dfly64.csv | head -20
nic0:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 73941, 9935377, 619, 111, 376
nic1:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 91979, 15047751, 638, 111, 371
nic2:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 110944, 19793500, 673, 111, 371
nic3:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 115826, 20774196, 708, 111, 411
nic4:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 121446, 24598190, 603, 198, 371
nic5:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 119880, 24013058, 603, 111, 371
nic6:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 119617, 23957537, 603, 111, 431
nic7:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 136591, 32116953, 603, 111, 371
nic8:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 170622, 51333116, 603, 197, 387
nic9:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 186753, 62059461, 603, 112, 379
nic10:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 170439, 51433653, 603, 111, 391
nic11:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 169842, 51051518, 603, 111, 371
nic12:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 188391, 62734343, 603, 198, 425
nic13:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 186742, 62051928, 603, 111, 371
nic14:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 186760, 62189826, 603, 111, 459
nic15:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 169829, 51057961, 603, 111, 384
nic16:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 171813, 51920143, 603, 198, 382
nic17:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 188681, 60303301, 603, 112, 396
nic18:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 186509, 59060635, 603, 111, 377
nic19:rtrLink:networkIF, packet_latency, , Accumulator, 2630646733, 0, 204697, 70941807, 603, 111, 407
stephenrajva@Stephens-MacBook-Air-4 dfly64 %
stephenrajva@Stephens-MacBook-Air-4 dfly64 % cd ~/Downloads/SST_Stimulator/MiniFE/DragonFly
stephenrajva@Stephens-MacBook-Air-4 DragonFly % nano dragonfly_comparison.csv
stephenrajva@Stephens-MacBook-Air-4 DragonFly % nano plot_dragonfly.py
stephenrajva@Stephens-MacBook-Air-4 DragonFly % python3 plot_dragonfly.py
Graphs generated
stephenrajva@Stephens-MacBook-Air-4 DragonFly % open sendbits.png
open packets.png
open latency.png
stephenrajva@Stephens-MacBook-Air-4 DragonFly %
stephenrajva@Stephens-MacBook-Air-4 DragonFly %
stephenrajva@Stephens-MacBook-Air-4 DragonFly % cd ~/Downloads/SST_Stimulator/MiniFE/DragonFly
stephenrajva@Stephens-MacBook-Air-4 DragonFly % nano extract_metrics.py
stephenrajva@Stephens-MacBook-Air-4 DragonFly % python3 extract_metrics.py
Traceback (most recent call last):
  File "/Users/stephenrajva/Downloads/SST_Stimulator/MiniFE/DragonFly/extract_metrics.py", line 23, in <module>
    latency = latency_df[10].mean()
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/util/_decorators.py", line 336, in wrapper
    return func(*args, **kwargs)
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/series.py", line 8113, in mean
    return NDFrame.mean(
           ~~~~~^
           self, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/generic.py", line 11831, in mean
    return self._stat_function(
           ~~~~~^
           "mean", nanops.nanmean, axis, skipna, numeric_only, **kwargs
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/generic.py", line 11785, in _stat_function
    return self._reduce(
           ~~~~~^
           func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/series.py", line 7480, in _reduce
    result = delegate._reduce(name, skipna=skipna, **kwargs)
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/arrays/string_.py", line 969, in _reduce

```

```
raise TypeError(f"Cannot perform reduction '{name}' with string dtype")
TypeError: Cannot perform reduction 'mean' with string dtype
stephenrajva@Stephens-MacBook-Air-4 DragonFly % nano extract_metrics.py
stephenrajva@Stephens-MacBook-Air-4 DragonFly % python3 extract_metrics.py
```

Processing DFLy4 ...

Processing DFLy16 ...

Processing DFLy64 ...

===== SUMMARY =====

	Topology	SendBits	Packets	Stalls	AvgLatency	AvgIdleTime	Throughput
0	DFly4	5847696	3462	0	263.50	2.408114e+08	22192.39
1	DFly16	14066216	20345	0	366.50	6.622161e+08	38379.85
2	DFly64	48972184	80627	0	383.38	4.777996e+08	127739.64

Saved: dragonfly_summary.csv

stephenrajva@Stephens-MacBook-Air-4 DragonFly %